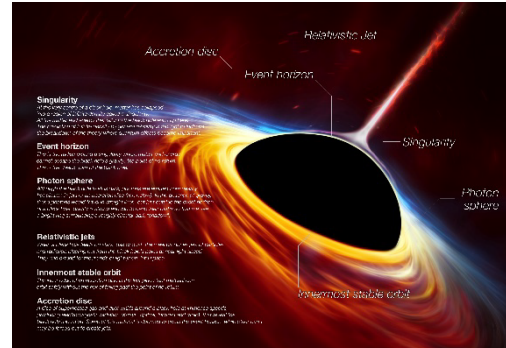


Why "We Don't Know" Is not an Equation.

What is a black hole?

A black hole is an object with an extremely dense region in space where gravity is so intense that not even light can escape. A black hole gets created when a dying star collapse into itself at the end of their life, creating black holes sometimes heavier than our sun by a factor of a million or more. Black holes are the space where GR (general relativity) and Quantum mechanics collide. On this paper I want to cover what the difference between density and mass is, the information paradox and the nature of the Einstein Rose bridge.



Clarifying the Singularity; Mass vs. Density

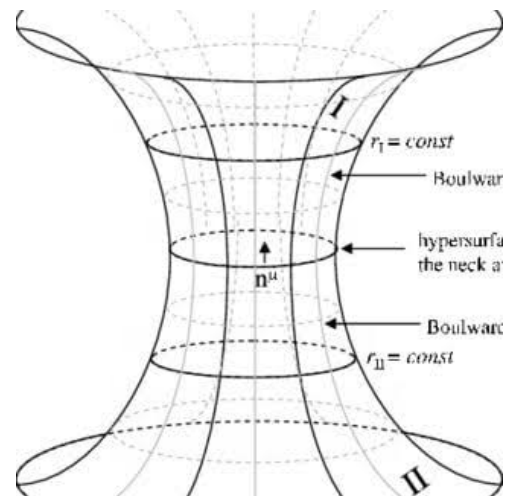
People often have theories that Black holes have infinite mass, which is not true. The fact that it is wrong is proven with the Schwarzschild metric.

$$ds^2 = -\left(1 - \frac{2GM}{c^2 r}\right) dt^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

M represents the finite mass of the original star that collapsed. If M were infinite, the metric would diverge everywhere in space, making gravity infinite at any distance r . However, the metric behaves completely normally for any distance greater than the Schwarzschild radius. This proves that the infinite divergence predicted by general relativity happens strictly at the centre, affecting the density, while the total mass M remains a fixed, finite constant. GR predicts a singularity with divergent density, not mass. Outside the event horizon the strength of the gravity remains finite and the gravity gets only altered by the original mass. The singularity happens exactly at $r = 0$. If the Distance r is any greater than the Schwarzschild radius $r_s = \frac{2GM}{c^2}$ the metric does behave normally. We can also further prove this with the Birkhoff's theorem; "Any spherically symmetric solution of the vacuum field equations must be locally static", which means that if the mass M remains constant the gravitational field stays the same, independent on whether for instance the Mass a tiny black hole or a massive star is. If two object have the same mass, they have the same gravitational field.

Einstein-Rosen bridge is non traversable.

Something that people also often claim is that the singularity of a black hole connects with a white hole safely, however the geometry of a standard black hole forbids this. While an Einstein-Rosen bridge created by a standard Schwarzschild black hole is possible, it is still dynamically unstable and pinches off faster than the speed of light. We can prove this by using multiple theorems.



Kruskal-Szekeres (KS) coordinates:

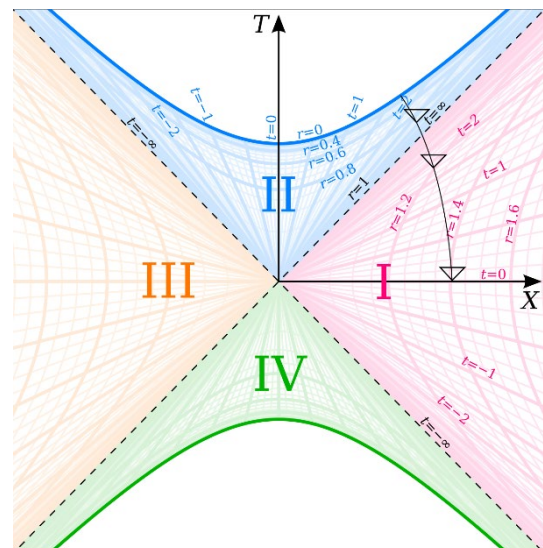
In short terms the Kruskal-Szekeres coordinates map allow us to map the entire spacetime geometry of a black hole. It is divided into four regions.

Region I: Our exterior universe.

Region II: The interior of a black hole.

Region III: A "parallel" exterior universe (or distant region of our own). Region III is mathematically interpreted as a "mirror" universe.

Region IV: The interior of a white hole.



A "Wormhole" is the bridge connecting Region I to Region III. When we map out the paths the physical particles and light can take (where null rays always travel at 45°) we see two physical barriers. To simulate a traverse, we will start at $t = 0$ in Region I. As we move forward in time (upwards) and inward (lefthand) we cross the dashed line at $r = 1$ and we enter Region II. Because we always must point 45° or more, our trajectory is forced to head straight up. No matter how hard someone might try to steer, they would always end up hitting the solid blue line $r = 0$ (The singularity). We cannot physically go below 45° because doing so would require us to move faster than the speed of light which physically is impossible. This explanation is a bit simplified for better understanding so take it with a grain of salt or consult this Wikipedia article:

https://en.wikipedia.org/wiki/Kruskal%E2%80%93Szekeres_coordinates

Topological Censorship Theorems

If we imagine a completely flat universe that contain two distant regions of space and are connected via a wormhole. An observer wants to start at infinity on one side $\mathcal{I}^-$, travel through the wormhole and safely exit into infinity on the other side $\mathcal{I}^+$. However Topological censorship proves that this is impossible since any signal sent from infinity to infinity must be topologically trivial and since a wormhole is a literal hole in the fabric of space, their path is tied to that hole (topologically non-trivial). In result this proves that it mathematically guarantees that the universe will dynamically censor this shortcut, forcing an event horizon to form and trapping the observer inside before they can ever cross.

"ANEC" (Averaged Null Energy Condition)

Standard physics state that gravity is purely attractive, the Mass and energy tell space how to curve, and normal mass/energy causes light rays to bend inward toward one another. This phenomenon is called focusing. ANEC says that energy must stay positive, which means that gravity must always focus light rays. Raychaudhuri's Equation is a fundamental equation in GR that tracks the expansion θ of a bundle of light rays. If $\frac{d\theta}{d\lambda}$ is negative the light rays are focusing together. For a wormhole you need a throat connecting two spaces. For light to enter and exit on the other side light must spread apart at the end of the throat which means that $\frac{d\theta}{d\lambda}$ must be positive. The only way to accomplish this is to have negative energy density. This requires exotic matter, a hypothetical form of matter that possesses negative mass-energy and violates standard energy conditions like ANEC. A standard black hole is made of normal mass which means that it obeys ANEC. To keep it open you would have to defy normal gravity as we know it by bending light rays and gravity using exotic matter.

Information Paradox & Unitary Violation.

Unitarity and the S-Matrix

If information would get randomized by Hawking radiation or leaves our universe, it violated the fundamental laws of quantum mechanics. The Bedrock of Quantum mechanics is unitarity. According to unitarity the total probability of all outcomes in the universe must equal exactly 100% and is governed by a unitarity operator U .

$$U^\dagger U = I$$

In short, the unitary is a mathematical property of transformations and operators ensuring that the total probability of a quantum state always sums to exactly 1. Time evolution

gets mapped using the Scattering Matrix (S -matrix). A unitary (S -matrix) guarantees that probabilities never become 95% or 105% for instance but are always 100% and that if you know the final state of a system you can run it mathematically backwards to see the original state of the system. Let us say we would throw a book into a Black hole. The black hole will eventually completely evaporate via Hawking radiation. Hawking radiation happens because pairs of virtual particles constantly pop into existence near a black hole's event horizon. Before they can destroy each other, one particle falls in while the other escapes. To keep energy balanced, the black hole absorbs negative energy, losing mass and slowly evaporating, while the escaping particle is seen as radiation. Hawking Radiation appears thermal and information free which means that it does not carry any information about the Book. Once the black hole completely vanishes, the space is left with a mixed quantum state, random thermal radiation.

Liouville's Theorem & Von Neumann

Liouville's Theorem states that the density of states in a system's phase stay constant over time. If we start with a specific volume, the Volume cannot shrink to zero. Von Neumann Equation governs this.

$$\frac{\partial \rho}{\partial t} = -\frac{i}{\hbar} [H, \rho]$$

Where ρ is the density matrix and H the Hamiltonian. This Equation relies on a unitary setup which means that it preserves the "purity" of a quantum state. Therefore, if you send information into a wormhole that pitches off or hits a singularity the information gets severed from out the density of our universes density matrix meaning the Von Neumann equation breaks down since the system is no longer unitary. A transition from a deterministic quantum evolution to an abrupt information losing dead-end is not possible.

No-Hiding Theorem

A counter argument might be that the information is not destroyed and just sitting inside the singularity or safely resting in Region III, but the No-Hiding theorem places strong restrictions on this idea. The theorem mathematically proves that information cannot just rest inside the remaining correlations, it must actively be transported to another part of the universe's active quantum state. As stated before, Region III is not accessible to us. If the information is trapped inside and eventually evaporates into nothing it cannot be transported back to our universe, meaning that from our universe quantum state perspective the information has vanished. Below is the Schrödinger's Equation that governs a state over time and is strictly reversible.

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$$

Epistemology

New theories do not erase old ones; they just reduce them using mathematical limits. When Einstein developed GR, he did not erase Newton's equations, he just built around it and strengthened and expanded the framework. For any new theory to be validated it must copy part of the old theory in the domains where the old one already works. Any future theory must satisfy this condition:

$$\lim_{\text{quantum effects} \rightarrow 0} (\text{Future Quantum Gravity}) = \text{General Relativity}$$

Conclusion

Only because “we don’t know yet” or because we don’t understand the advanced physics beyond our current cognitive understanding doesn’t mean that anything is possible, or we can simply state the theories as only theoretical and ignorable; any theory relies on proven physics and is based off other theory’s. Arguing that theoretical thing could be possible since they are theoretical is not true unless you can scientifically prove a part of them; Just like the Einstein-Rose bridge equation which does prove that a wormhole is possible but also proves that it can’t be traversed since it’s dynamically unstable. It is important to mention This should not discouraged people from making up own theories or questioning the system but should also think about what their theory would mean in physics.

Sources

Used to get Equations and Pictures

<https://www.space.com/15421-black-holes-facts-formation-discovery-sdcmp.html>

https://www.researchgate.net/publication/260056640_The_Schwarzschild_black_hole

https://www.researchgate.net/publication/1977725_Topological_Censorship_and_Higher_Genus_Black_Holes

chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://sites.pitt.edu/~jdnorton/teaching/HPS_0410/chapters/black_holes_Einstein/Kruskal_Szekeres.pdf?utm_source=chatgpt.com